# Current: charge flow rate

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Calculate current as charge transferred per time.

**SUCCESS CRITERIA - I can...**

1. Use I=q/t with seconds and coulombs.
2. Describe conventional current direction.
3. Explain steady charge conservation at a junction.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “current” and “charge” to explain your model or result. Sentence starters are optional.

**current**: rate of charge flow

**charge**: a property responsible for electric interactions

**interval**: the time or range between two stated values

## 01 Warm-up: bring back an earlier idea

Keep charge separation fixed and multiply one charge magnitude by 3.
**Compare** electric-force magnitude.

**Word help:** separation = distance between the specified objects or centers

**Sentence starter:** Both ... . They differ in ... because ... .

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $I = \Delta q/\Delta t$.
Current is charge per time; 1 A = 1 C/s. Conventional current follows positive charge motion.

**Worked reasoning:** If 12 C pass a point in 4 s, average current is $I=Q/t=3$ A. Charge circulates; a resistor does not consume charge.

**Your turn: Draw** a complete loop and label equal current before and after one resistor in steady state.

**Sentence starter:** The ... in my model represents ... . Its label shows ... .




Label component values, current and voltage as needed.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 03 Read and interpret the model

**Draw** conventional current around the closed circuit, outside the battery from + toward -. What happens if a wire is disconnected?

**Sentence starter:** The … in my model represents … . Its label shows … .



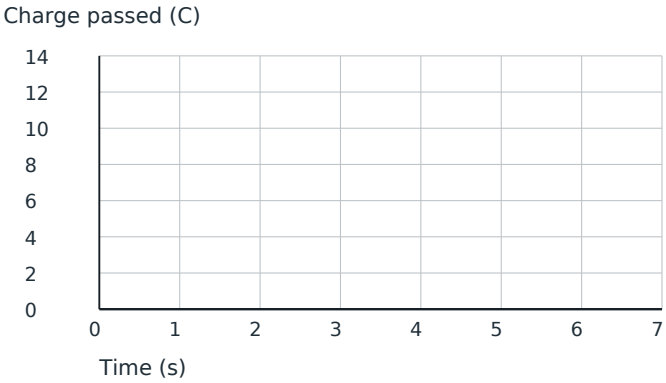

Circuit symbols: long/short lines = battery; rectangles = resistors.

**Marzano check - circle:** 1 2 3

## 04 Apply the physics

During a circuit test, 12 C passes one point in a wire over 7 s. Plot cumulative charge against time for a constant current.
**Calculate** current from the graph's slope.

**Sentence starter:** I found … [unit]. This value means … in this situation.




**Marzano check - circle:** 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Explain your reasoning

At a junction, 5 A enters and 2 A leaves along one branch.
**Find** the other outgoing current in steady operation.
**Explain** the conservation rule.

**Sentence starter:** My claim is … . The diagram/data show …, so … .

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

A siren emits 600 Hz steadily while approaching a stationary listener.
**Compare** observed and emitted frequency and explain.
**Use** wavefront spacing to explain the frequency change.

**Sentence starter:** Both … . They differ in … because … .

Source moves right. Add wavefronts.



**Marzano check - circle:** 1 2 3

## 07 Exit check

A current of 2 A flows for 3 s. How much charge passes?

**Sentence starter:** I notice … . This supports … because … .




Label component values, current and voltage as needed.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

Construct two charge -time pairs that both give 2 A.
**Explain** why different total charges can give the same current.

**Sentence starter:** Both … . They differ in … because … .




Label component values, current and voltage as needed.

**Marzano check - circle:** 1 2 3